



Ricardo G. @ AC

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Key takeaways

What's a computer 101

"A computer once beat me at chess, but it was no match for me at kickboxing"

00



Wee *bit* of history

Ancient aliens: The origins of lobster fingers

01



// == 2 == 10

Because you know I'm all about that base... 1, 10, 2...

10



11

Bonds and broken bonds

What goes around, goes around,
goes around
Comes all the way back around



100

Continuously Analog

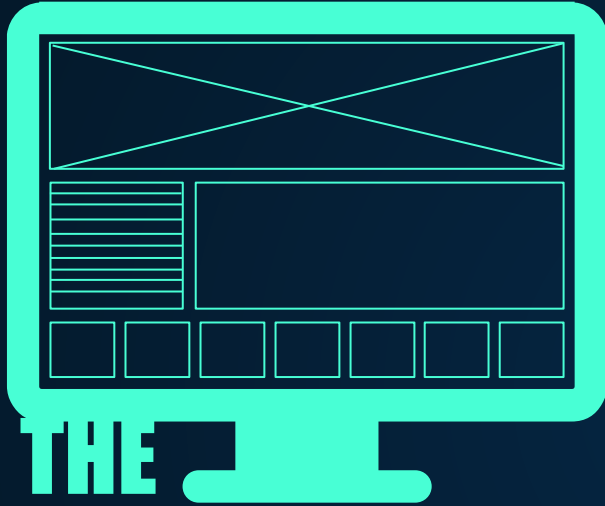
Analogies are like donuts on a
trampoline:
They don't always work.



101

Digital is where it's at

Relax, it's only ONES and ZEROS !



THE COMPUTER



101...?

- An electronic machine that can store, organize and find information, do processes with numbers and other data, and control other machines.
- A computer is a machine that can be programmed to carry out sequences of arithmetic or logical operations (computation) automatically.
- A programmable usually electronic device that can store, retrieve, and process data

History: 10 fingers alone won't cut it



1st there was 1

One finger, one item. Five fingers, five items. The Unary System.

Let there be 2 or 6 or 10 or 60

One finger, one item. One hand, five items. That's more like it!

And then was a whole lot of nothing (0)

Your value comes not from who you are, but where you are.

Numeral systems helpdesk support



base-2

Aka: binary numeral system

**How to get those
calculations going**

+ table	0	1
0	0	1
1	1	0 (carry 1)

x table	0	1
0	0	0
1	0	1

Example					1010	
					x	1011
			1	0	1	0
		1	0	1	0	X
	0	0	0	0	X	X
1	0	1	0	X	X	X
1	1	0	1	1	1	0

Numeral systems helpdesk support



base-2

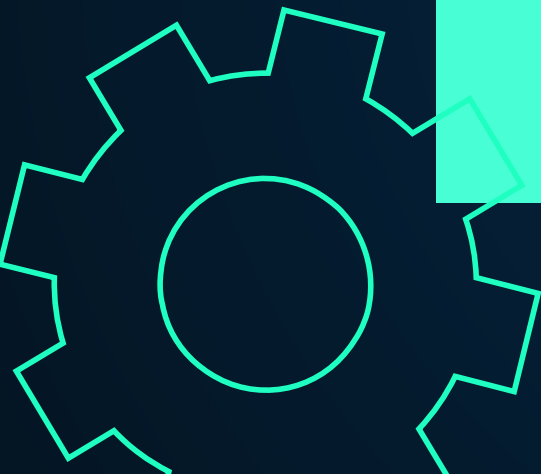
Aka: binary numeral system

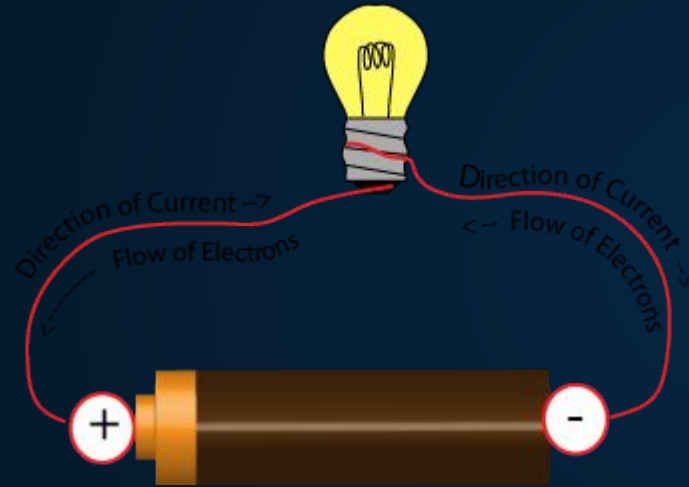
Conversion table	And so forth...	2^3	2^2	2^1	2^0
Same as	(...)	8	4	2	1
Binary Number		1	0	1	0
Decimal Number	Sum = 10	$+8 \times 1 = 8$	$+4 \times 0 = 0$	$+2 \times 1 = 2$	$+1 \times 0 = 0$

How to know what these calculations are doing



"When it's time, its time... and it's
time."





MAGNET(ISM)!

Positive Atoms have less electrons and vice-versa.

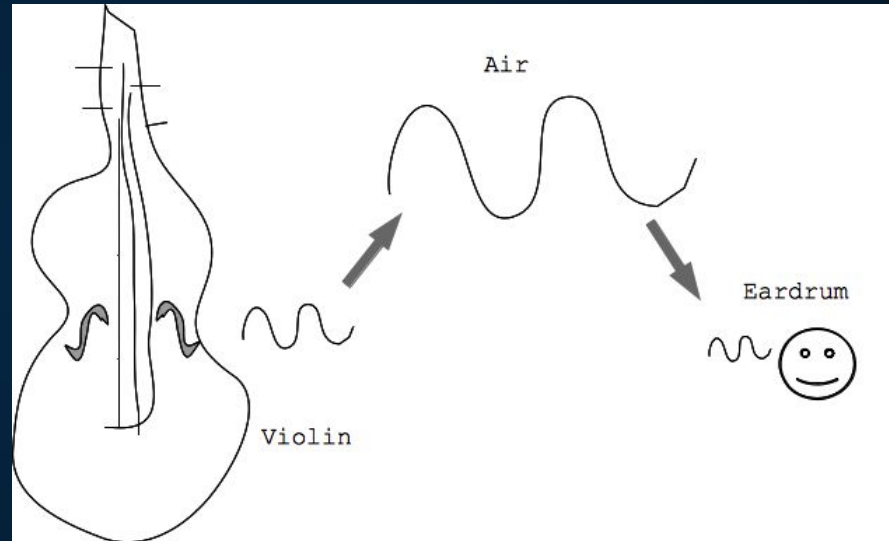
Analog



Continuous over time

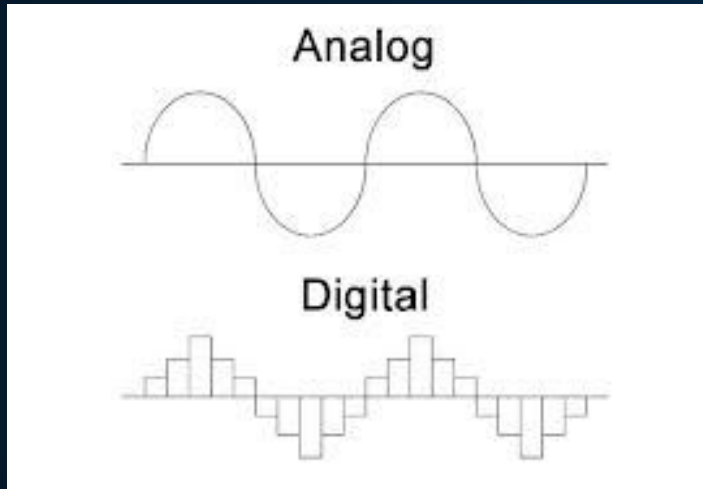
Can be split infinitely

Vulnerable to "noise"





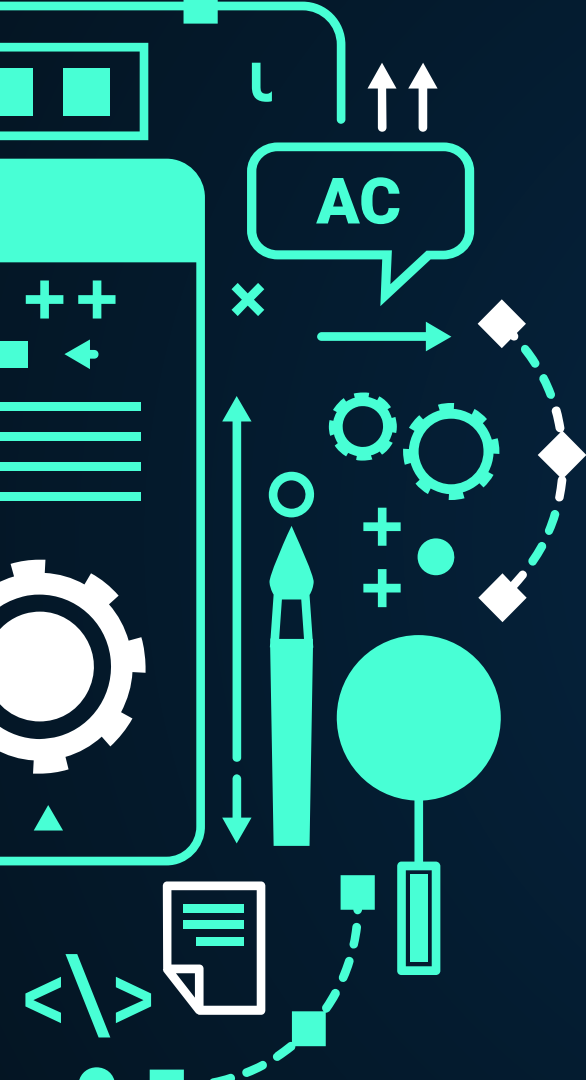
DIGITAL



SAMPLED

IMPERVIOUS TO NOISE

RELIABILITY



THANKS!

Does anyone have any question?